Wrong or deliberate patient clicks

How AIMScribe v3 keeps every consultation under the right patient when a doctor opens another patient's profile - by mistake or on purpose - at the start, in the middle, or between consultations. Follow-up to the design review; updated 20 October 2026 for recorder 3.5.0.

The problem

A recording starts when the doctor opens a patient in CMED (API 1). Doctors sometimes open the wrong patient first and correct it within seconds; sometimes open another patient's file on purpose for a moment; and sometimes finish a follow-up visit with no new prescription, so no 'consultation finished' signal (API 3a) ever comes before the next patient. In every case the recording must end up under the patient who was actually consulted - and no audio may be lost or deleted.

Constraint	Consequence for the design
CMED's code is final	Only the messages CMED already sends can be used; every reply must be one their code already handles
Only API 1 (patient opened) and API 3a (prescription built) reach the recorder	The recorder cannot see API 2 or 3b; only our server sees all four
At the moment of a click, intent is unknowable	A wrong pick, a glance and the next patient after a follow-up look identical when they happen

The principle: never refuse, never stop, never delete - decide later

Where	What it does
Recorder, live	Before a prescription, every click on any patient is accepted (RECORDING_STARTED with the running recording) and the recording carries on, so a glance never cuts a consultation. Each click and each prescription is signed into the recording with its exact second of audio. After a prescription, the next different patient closes the recording and starts a new one.
Server, when the recording closes	Reads the order of clicks and prescriptions, plus every API 2 and 3b, and applies five rules (below) to decide whose audio is whose. A 3b that arrives later makes it decide again.
Archive	Cuts the audio at the exact sample of a click into one file per patient. The signed recording is never altered: the files rejoin byte for byte to it, which is checked.

The five rules

Rule	Evidence	Verdict
R1	A prescription (3a or 3b) for a patient	That patient was consulted; their part is never merged away
R2	The doctor went back to an earlier patient (reopened them, or built their prescription)	What lay between was a glance: it belongs to that patient
R3	A patient open under 60 s, never returned to, never prescribed	A wrong pick or stray click: it belongs to the next patient
R4	A patient open 60 s or more, no prescription, no return	A real consultation without a prescription (follow-up): its own file
R5	Anything still unclear	Kept, labelled by the best rule, and flagged for a person on the dashboard's Needs review panel - never deleted

CASES

Every case, at the start, in the middle and between consultations

What the doctor did, what arrives, what the recorder does live, and the files the archive writes.

When	What the doctor did	What arrives	Recorder, live	Files written
Start	Opens Rahim by mistake, opens Karim 8 s later, writes Karim's prescription	API 1 Rahim, API 1 Karim, API 2 for both, 3a/3b Karim	One recording; both clicks accepted	**One file, Karim** (R3+R1). Rahim's API 2 kept, with no audio
Start	Opens the right patient, deliberately opens another for 10 s, goes back, prescribes	API 1 A, API 1 B, API 1 A again, 3a/3b A	One recording, nothing cut	**One file, A** (R2). B's API 2 kept, no audio
Start	Opens A, opens B within a minute and stays with B, prescribes B	API 1 A, API 1 B, 3a/3b B	One recording	**One file, B** (R3+R1): a first minute that was never returned to is a correction
Middle	Mid-consultation glance at another file, then back to the patient and prescribes	API 1 B mid-way, then 3a/3b A	Recording carries on	**One file, A** (R2): building A's prescription counts as going back
Between	Follow-up with no prescription (Sakib), then the next patient (Salma), who is prescribed	API 1 Sakib, API 1 Salma, 3a/3b Salma	One recording until the next click after Salma's prescription	**Two files**: Sakib from the start to Salma's click (R4), Salma from there (R1)
Between	Two or three follow-ups in a row, then a prescribed patient	API 1 for each, 3a/3b for the last	One recording	**One file per patient**, each cut at the next patient's click (R4 ... R1)
Between	Accidental tap on another patient after the prescription	3a A, API 1 B, then API 1 C	A closes at the tap; a new recording starts	A stays A; the short tap on B is merged into C (R3)
End	Last patient, prescription, nobody clicks again	3a A only	Closes itself 2 min after the prescription (up to 10 s more, at a pause)	**One file, A**; the doctor's own time is not recorded
End	Last patient, a follow-up with no prescription, no further click	API 1 only	Closes after 15 min without CMED activity, at Stop, or cleanly at Windows shutdown	**One file** (R4)
Any	The same patient clicked twice or the page reloaded	API 1 again	Same recording, no extra marker	No change

Four of them, drawn to scale

1. Wrong pick at the start: Rahim by mistake, Karim 8 s later

2. Deliberate click on another file, then back

3. Glance in the middle of a long consultation

4. Follow-up with no prescription, then the next patient (live run)

Two API 2s for one recording

Each click CMED accepts brings that patient's API 2 to our server. API 2 means only 'this profile was opened' - it cannot tell a consultation from a glance, and neither is wrong data: both are genuine records about real patients. So no API 2 is ever deleted or trusted on its own. The recorder asks the server for a permission for each clicked patient, which ties that patient's API 2 and 3b to this recording. The rules above decide which of them gets audio; the others stay as 'opened, not consulted'. A late record for a patient whose part was moved elsewhere is never attached to someone else's file.

What a click can never do (recorder 3.5.0 and the current server)
Erase a consultation. Nothing the page sends can: a refusal sent over the socket is recorded as an ordinary stop and the audio is kept. Only the doctor's on-screen 'patient refused' control erases, and the recorder signs that refusal with the PC's own key; the server then removes the audio, the clinical record, the archive file and its cloud copy.
Put one patient's audio in another's file. Each recording has its own clip writer: even with a full disk, a clip that waited is written into the recording it was captured in, never the next patient's. Every clip's signed record names that clip, so the server refuses one filed elsewhere.
Reach another doctor's recorder. With two Windows users signed in, a page in the other user's session is refused.

Evidence

What	Result
Live, real recorder, real server and archive (16 October)	Wrong pick: one file, Karim, R3+R1. Follow-up: Sakib 0-70.1 s (R4) and Salma from 70.1 s (R1). Counselling limit closed the recording 120.2 s after the prescription. A pause during a recording kept the cut exact
Integrity of the cut	Sakib's and Salma's files rejoined to the signed recording's SHA-256 exactly
Server rules	backend tests/test_labelling.py: every case above plus rapid clicks, double clicks, out-of-order events, a late 3b changing a verdict
Recorder	tests/test_markers.py and test_channel_a.py: replies, markers, both time limits, the grace at a pause
Archive	archive_worker test_parts.py: cuts exact to the sample, parts rejoin, interrupted runs resume, re-cuts refused if the hash differs
Against CMED's finished software	Not yet: one session with tools/cmed_build_session.py will show their page's real messages

What remains, and how it is caught

Case	Why it cannot be settled automatically	What happens
A follow-up shorter than 60 s	Looks exactly like a wrong pick	Merged into the next patient; flagged for review
A glance with no return signal and no prescription	Nothing says it was a glance	Kept as its own part; flagged for review
A phone call in the room during a follow-up	Speech is speech; no software can tell a call from a consultation	The doctor's Pause button; the 15-minute limit caps it
CMED's page not sending API 2 for a click	Their code, not ours	The part is archived unconfirmed and kept; checked in the CMED session

To check it yourself: the CMED page's manual-test cards 1-8 (http://localhost:3000/manual-test) and Part 4A of the local manual.